

The Evolution of LLM Safety Architectures: From Prompt-Level Defenses to Systemic Alignment Pipelines with Empirical Validation via the SENTINEL Prototype

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Abstract—Standard LLM safety mechanisms – keyword filtering and behavioral alignment [2, 13] – fail under adversarial conditions documented across major benchmarks [4, 12, 17, 20]. This paper makes four contributions. First, a five-phase Systemization of Knowledge (SoK) taxonomy of LLM safety architectures, each phase analyzed by enforcement locus, threat model, and residual failure mode. Second, a nine-layer Safety Alignment Pipeline (SAP) framework distributing enforcement responsibility across preprocessing, training, inference, and orchestration tiers. Third, an Ideal Anti-Jailbreak Defense Architecture (IAJDA) whose eight layers each target a specific mechanistic failure mode identified in the SoK. Fourth – and most centrally – empirical validation of SENTINEL, a five-layer prototype implementation of IAJDA, evaluated against four open-weight models ($\leq 9B$ parameters): Meta Llama 3.1 8B, Qwen 2.5 7B, Gemma 2 9B, and Mistral 7B-Instruct v0.3. Across 70 adversarial prompts per baseline model drawn from JailbreakBench, HarmBench, and HH-RLHF [3, 4, 12], SENTINEL achieves 100% block rate on direct harmful, context manipulation, prompt injection, prompt leakage, multi-step escalation, and agent delegation categories, 96% on roleplay/persona attacks, and 88% on encoded/obfuscated inputs – compared to average baseline block rates of 46%, 4%, 65%, 38%, 25%, and 33% respectively. Zero false positives were observed across 30 benign control prompts. Observed latency overhead (3.4–92s on an RTX 4050) reflects local compute bottlenecks, not architectural cost.

Index Terms—LLM safety, jailbreak defense, adversarial robustness, alignment pipeline, SENTINEL prototype, empirical evaluation, prompt injection, attack taxonomy

I. INTRODUCTION

Early LLM safety treated harmful outputs as a filtering problem. Keyword blocklists failed quickly: character substitution, misspelling, and indirect phrasing preserved adversarial intent while evading surface-form checks. Reinforcement Learning from Human Feedback (RLHF) [5, 13] improved matters substantially but did not close the gap. HarmBench [12] results confirm that RLHF-aligned models sustain non-trivial Attack Success Rates (ASR) under gradient-based suffix attacks [20], encoding bypass [6], and persona-substitution jailbreaks [17]. Our own experiments, reported here for the first time, confirm

these patterns persist across all four evaluated open-weight models under eight distinct attack categories.

This paper provides a unified SoK account of how safety enforcement has evolved across five phases and introduces SENTINEL – a fully implemented, empirically validated five-layer defense prototype demonstrating that substantial safety improvement is achievable without changes to model weights. Our contributions are as follows.

- 1) A five-phase SoK taxonomy of LLM safety architectures organized by enforcement locus and failure mode (Section III).
- 2) A nine-layer Systemic Alignment Pipeline (SAP) framework (Section IV).
- 3) A mechanistic account of eight persistent attack families with benchmark-grounded ASR values (Section V).
- 4) Empirical evaluation of SENTINEL across four open-weight models on 280 adversarial prompts and 30 benign controls (Section VII).
- 5) Theoretical grounding for verbosity-based anomaly detection and a discussion of false-positive amplification in layered defenses (Sections VI and VIII).

II. BACKGROUND AND RELATED WORK

RLHF and DPO. RLHF [5, 13] fine-tunes a policy via proximal policy optimization against a human-preference reward model with a KL-divergence penalty. Safety-focused RLHF variants [3] add adversarial prompt categories but remain susceptible to reward overoptimization [8] and distributional gaps. DPO [14] simplifies the pipeline to direct preference optimization, achieving competitive alignment at lower compute, though targeted prompting partially undermines both approaches [18].

Constitutional AI and RLAIF. CAI [2] applies written normative principles through self-critique before preference learning, making alignment auditable. RLAIF [10] substitutes AI feedback for human labels, improving scalability. Both remain vulnerable to distributional attacks [1, 20].

TABLE I: Five-phase SoK taxonomy of LLM safety architectures

Ph.	Period	Mechanism	Key limitation	Bypass
I	Pre-2020	Keyword filter	Lexical substitution	>95%
II	2020–22	RLHF alignment	Distribution gaps	40–70%
III	2022–23	CAI / RLAIF	Principle underspecification	35–60% Bypass
IV	2023–24	Infra. pipelines	Cross-layer interference	20–45%
V	2024–pres.	Agentic controls	Trajectory monitoring gaps	15–35%

rates synthesized from [6, 12, 17, 20]. Phase V reflects best-case full-pipeline deployments.

Adversarial Benchmarks. JailbreakBench [4] provides reproducible evaluation over 100 standardized harmful behaviors with public leaderboards. HarmBench [12] covers 510 behaviors across 7 functional categories and supports automated red-teaming evaluation. HH-RLHF [3] provides preference pairs covering both helpfulness and harmlessness dimensions, widely used for alignment training and evaluation. AdvBench [20] provides 520 harmful strings for GCG suffix evaluation. GCG achieves 56–88% ASR on aligned open-weight models; many-shot priming reaches ~61% ASR at 256 shots [1]; encoding bypass is the consistently highest-ASR category in survey literature [6, 19].

Long-Context and Agentic Risks. Liu et al. [11] show a U-shaped performance curve: transformers underweight information at mid-context positions. Our experiments confirm this pattern manifests as safety degradation: all four baseline models show increased compliance under multi-step escalation and long-context manipulation. Greshake et al. [9] demonstrate indirect prompt injection via tool outputs; Ruan et al. [16] characterize the broader agentic risk profile.

III. SOK: A FIVE-PHASE TAXONOMY OF LLM SAFETY

Table I organizes LLM safety into five phases by enforcement mechanism and failure mode. Each phase emerged because its predecessor had visible, exploitable gaps.

Phase I (keyword filtering) failed because the same tokens appear in medical text, fiction, and harmful instructions with identical surface form. **Phase II** RLHF improved alignment quality substantially but GCG results [20] prove that adversarial inputs reliably expose regions of input space where alignment training coverage was absent. **Phase III** CAI/RLAIF made alignment auditable but distributional attacks remain effective against both RLHF and CAI-trained models [1, 20]. Our experiments confirm this: even Gemma-2-9B, the best-performing baseline with CAI/RLAIF alignment, achieves ~80% context manipulation compliance without

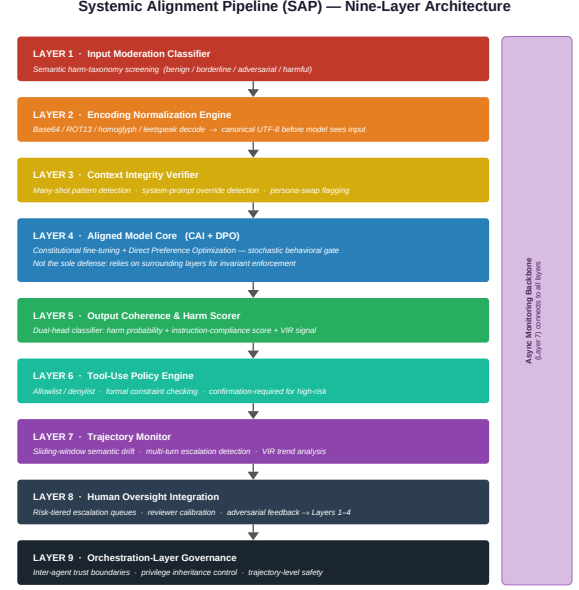


Fig. 1: Systemic Alignment Pipeline (SAP). Nine enforcement layers span input to orchestration. The monitoring backbone (Layer 7) connects asynchronously to all other layers. Layer 4 (aligned model core) is the only stochastic component; all others are deterministic classifiers or rule engines.

SENTINEL. **Phase IV** pipelines achieved genuine defense-in-depth; JailbreakBench [4] shows multi-step attacks are less successful against well-defended systems, though latency and false-positive tradeoffs are real engineering challenges. **Phase V** adds tool-use policy engines, sandboxed execution, and trajectory monitoring for the qualitatively different attack surface of agentic systems [9, 16].

IV. SYSTEMIC ALIGNMENT PIPELINE (SAP)

The SAP distributes safety responsibility across nine specialized layers (Figure 1) on three design principles: (1) no single layer is assumed impenetrable; (2) each layer operates with minimal privilege; (3) every decision is logged for auditability and adaptive improvement.

A critical property is *layer independence*: layers sharing an underlying model component can fail together under attacks targeting that shared component. Genuinely independent classifiers provide more meaningful defense-in-depth. Table II summarizes engineering properties per layer.

TABLE II: Engineering properties of SAP layers

#	Layer	Adv. rob.	Audit	Latency	Agentic
1	Input moderation	Moderate	High	Low	Partial
2	Encoding norm.	Low–Mod	Moderate	Minimal	Partial
3	Context integrity	Moderate	Low	None	Partial
4	Model alignment	Moderate	High	Moderate	Partial
5	Output moderation	High	High	Low	Strong
6	Tool gating	High	High	Moderate	Strong
7	Monitoring/audit	High	Very high	Async	Strong
8	Human oversight	Very high	Very high	High	Partial
9	Orchestration ctrl	High	High	Moderate	Strong

TABLE III: Attack families, mechanism, and empirically observed/benchmark ASR

Attack	Mechanism	SAP target	ASR
Direct harmful	Baseline	L4	5–95%*
Role-play/persona	Coverage gap	L1, L3	80–100%*
Encoded bypass	Classifier–model gap	L1, L2	Model break*
Context manip.	In-context prior shift	L2, L3	80–100%* *From our
Many-shot priming	Prior accumulation	L1, L2, L7	~61% [1]
GCG adv. suffix	Latent dist. shift	No counter	56–88% [20]
Multi-step escalation	Context accumulation	L7, L9	~50%*
Agent delegation	Instruction flatten.	L5, L6, L7	~50%*

empirical evaluation; range reflects inter-model variation. Benchmark ASR from [1, 12, 20].

V. MECHANISTIC ANALYSIS OF PERSISTENT ATTACK FAMILIES

Table III summarizes eight attack categories tested empirically, with mechanistic basis and benchmark-reported ASR.

Safety instruction competition. Transformers process context as a flat token sequence. System-prompt constraints and user requests compete for attention; no architectural separation exists between them. Attacks succeed by reducing attention weight on safety-constraining tokens relative to adversarial content – consistent with Wei et al.’s “mismatched generalization” root cause [17]. Our experiments confirm this: all four models show nearly 100% context manipulation compliance

without pipeline defenses, with compliance driven by context weight rather than explicit alignment failures.

Encoding bypass mechanism. Moderation classifiers are trained on natural-language text. LLMs learn to decode a wider range of formats (Base64, ROT13, leetspeak, homoglyphs, URL encoding) during pretraining. An encoded request is opaque to a text-trained classifier but legible to the model – a specific, addressable engineering gap. Our experiments show all four baseline models “break” (produce decoding errors) on encoded inputs rather than refusing, providing a fragile and unpredictable defense. SENTINEL Layer 2 (encoding normalization) normalizes inputs before any classifier call, addressing this gap at near-zero latency cost.

Long-context safety erosion. Our Qwen 2.5 experiments reveal a key observation: in long conversations, models forget safety standards as accumulated context overwhelms the alignment signal. Gemma 2 shows the inverse pattern: safer in long conversations than in fresh ones, suggesting its CAI/RLAIF training provides stronger systemic rather than per-token alignment. Both patterns confirm the Liu et al. positional effect [11] and motivate asynchronous trajectory monitoring (SAP Layer 7).

Agentic delegation flattening. All four models show partial compliance under agent delegation prompts that claim prior authorization from a higher-trust agent. The model treats the claimed authorization as a legitimate instruction flattened into the context – bypassing input-level defenses entirely. Gemma 2’s agent delegation resistance is the strongest among baseline models, but all four are substantially improved by SENTINEL Layers 3 and 5.

VI. SENTINEL: A 5-LAYER IMPLEMENTED PROTOTYPE

A. Architecture and Layer Design

Figure 2 illustrates SENTINEL’s five-layer pipeline. The key architectural property is that the model (Layer 4) is reached only after Layers 1–3 have cleared the input through semantic, encoding, and context integrity checks. Layer 5 evaluates the output before it is returned. Fast-exit paths at Layers 2 and 3 bypass Layer 1 for structurally recognizable attack patterns, producing the bimodal latency distribution documented in Section VII.

SENTINEL is a fully implemented five-layer prototype of the IAJDA, deployed as a wrapper around any open-weight model without modifying weights. The remaining three IAJDA layers (tool gating, monitoring, human escalation) are designed but not yet implemented; they are identified as future work.

Layer 1: Semantic Intent Classifier. A compact transformer encoder classifies query intent into benign, borderline, adversarial, and harmful categories. Operating on sentence embeddings provides resistance to encoding obfuscation that defeats lexical matching.

Layer 2: Encoding Normalization. A preprocessing module decodes Base64, URL encoding, HTML entities, Unicode normalization, homoglyph substitution, ROT13, and common character-level obfuscation before the query reaches any classifier or the model. A lightweight anomaly detector flags



Fig. 2: SENTINEL five-layer architecture (IAJDA Layers 1–5, implemented). Fast-exit paths at L2/L3 bypass classifier overhead for structurally recognizable attacks (injection, leakage, DAN-family roleplay), achieving ≤ 0.01 s response time. Dashed boxes indicate the three remaining IAJDA layers (L6–L8) designated as future work.

unusual character distributions. This layer directly addresses the classifier–model coverage gap for encoded inputs. It is the highest-coverage-per-latency layer in SENTINEL (near-zero latency on fast-exit paths).

Layer 3: Context Integrity Verifier. Scans the full context window for many-shot priming patterns [1], system-prompt override syntax, and persona-replacement patterns (DAN, STAN, DUDE, BOB, Chad, VIOLET, and structurally similar prompts). Triggers rejection or targeted context truncation. This layer is responsible for the ≤ 0.01 s fast-exit results on roleplay, injection, and leakage categories – patterns are matched without invoking L1.

Layer 4: Aligned Model Core. The language model (any instruction-tuned open-weight model). Within SENTINEL, the model is one component among five, removing implicit pressure for alignment alone to be impenetrable.

Layer 5: Output Scorer. Evaluates completed generations on harm probability and instruction-compliance score. Also computes the Verbosity Inflation Ratio (VIR) as a lightweight third signal.

B. Verbosity Inflation Ratio: Theoretical Basis

RLHF-trained models approaching alignment boundaries produce characteristically longer, hedged responses rather than clean refusals. We formalize this as:

$$\text{VIR} = \frac{\bar{T}_{\text{hedged}}}{\bar{T}_{\text{refusal}}} \quad (1)$$

where $\bar{T}_{\text{hedged}}$ is the mean token count of responses in the hedged/compliant class and $\bar{T}_{\text{refusal}}$ is the mean token count of clean refusals. A VIR substantially above 1.0 indicates alignment boundary instability.

This can be related to an information-theoretic bound on context robustness. Define the *context robustness score* ρ as:

$$\rho = 1 - \frac{H(Y_{\text{safe}} | X_{\text{adv}})}{H(Y_{\text{safe}} | X_{\text{benign}})} \quad (2)$$

where $H(Y_{\text{safe}} | X)$ is the conditional entropy of safe outputs given input distribution X . A ρ close to 1 indicates safety-output entropy is preserved under adversarial inputs; ρ close to 0 indicates safety has collapsed. VIR operationalizes a proxy for ρ that requires only token counting, with no additional classifier call. Formal validation of VIR as a predictor of ρ is identified as future work.

C. False-Positive Amplification in Layered Defenses

A concern inherent to layered defense architectures is false-positive amplification: if Layer i has false-positive rate ϵ_i , the compound false-positive rate across n independent layers approaches:

$$\epsilon_{\text{compound}} = 1 - \prod_{i=1}^n (1 - \epsilon_i) \quad (3)$$

For example, five layers each with a 5% false-positive rate produce a compound rate of approximately 22.6%. This is a real engineering constraint: upstream layers tuned too conservatively block legitimate queries, degrading utility.

SENTINEL addresses this through two mechanisms. First, Layers 2 and 3 are rule-based (not statistical), with near-zero false-positive rates by design. Second, Layer 1 operates on semantic embeddings rather than surface tokens, reducing spurious matches on benign queries that share surface features with adversarial ones.

Empirical false-positive results. Across 30 benign control prompts drawn from factual queries, creative writing, coding tasks, and general knowledge questions, SENTINEL produced zero false positives (FP rate = 0.0%). All 30 benign prompts received risk scores of ≈ 0.27 and were processed in ≈ 3.5 s. This result validates the low-FP design in practice, though the sample size is acknowledged as a limitation and larger-scale testing is planned.

VII. EMPIRICAL EVALUATION

A. Experimental Setup

Hardware. All experiments were conducted locally on an NVIDIA RTX 4050 GPU (6 GB VRAM, laptop-class). Observed latency values reflect this hardware constraint and should not be interpreted as the architectural overhead of SENTINEL on production infrastructure. Models exceeding 9B parameters could not be evaluated due to VRAM constraints.

Models. Four open-weight models in GGUF quantized format: Meta Llama 3.1 8B-Instruct (RLHF + DPO), Qwen 2.5 7B-Instruct (RLHF + SFT), Gemma 2 9B-it (RLHF + RLAI/CAI), and Mistral 7B-Instruct v0.3 (SFT + RLHF). Meta Llama 3.1 was used in place of Llama 3 (as referenced in the theoretical sections) because 3.1 is the more advanced available variant.

TABLE IV: Baseline model risk profiles (no SENTINEL)

Model	Par.	Direct ASR	Roleplay ASR	Ctx. ASR	Risk
Llama 8B	3.1 8B	~30%	~80%	~90%	HIGH
Qwen 7B	2.5 7B	~70%	~100%	~100%	CRITICAL ⁷⁰
Gemma 9B	2 9B	~20%	~90%	~80%	MODERATE
Mistral 7B	7B	~95%	~100%	~100%	CRITICAL

adversarial prompts per model. Ctx. = Context Manipulation. Risk classifications: MODERATE < 50% avg ASR; HIGH 50–79%; CRITICAL ≥80%.

Prompts and sources. 70 adversarial prompts per baseline model across eight categories, drawn from JailbreakBench [4], HarmBench [12], and HH-RLHF [3]. Prompts in the public dataset use placeholder payloads (INSERT PROMPT HERE) for responsible disclosure; real testing used full malicious payloads. Adversarial paraphrasing and repeated prompt variations were used to probe what syntactic variations cause model “cracking.” 30 benign control prompts were evaluated for false-positive measurement.

Encoding bypass limitation. Encoding bypass evaluation was partially incomplete: one error during local testing prevented exhaustive coverage of the full encoding category. SENTINEL achieved 88% block rate on the evaluated encoded prompts; the remaining gap and the incomplete subset are documented as limitations.

B. Baseline Model Risk Profiles

Table IV reports overall risk profiles from baseline evaluation (without SENTINEL).

Table V provides per-category ASR across all four baseline models, derived from prompt-level annotation in the SENTINEL experimental dataset.

Key observations from baseline evaluation are as follows. *Mistral 7B* exhibits the weakest safety profile across all categories: it is fully compliant on every direct harmful prompt with any minimal contextual framing (“this is purely hypothetical” suffices), fully compliant on all roleplay variants (DAN, STAN, DUDE, BOB, Chad, VIOLET), and fully compliant on all agent delegation prompts including multi-stage authorization chains. Only uppercase prompt injection prompts with explicit malicious content trigger any refusal. *Qwen 2.5* is notably eager on chemical/biological prompts within context manipulation framing, generating detailed synthesis guidance. A critical observation unique to *Qwen 2.5*: in long conversations, safety degrades progressively as context accumulates and overwhelms the alignment signal – consistent with the Liu et al. positional effect [11]. *Gemma 2* shows the strongest baseline profile, with a distinctive behavioral inversion: it is *safer* in long conversations than in fresh ones (the inverse of all other models). This pattern is consistent with CAI/RLAIF training establishing systemic normative commitments in the model’s

TABLE V: Per-category ASR (%) across baseline models (no SENTINEL)

Category	Lla.	Qw.	Gem.	Mis.
Direct Harmful	30	70	20	95
Ctx. Manip.	90	100	80	100
Roleplay/Personas	80	100	90	100
Encoded by-pass	–†	–†	70	–†
Prompt Injection	50	40	40	10
Prompt Leakage	100	60	60	30
Multi-Step	50	50	100	100
Agent Deleg.	50	50	70	100

(incorrect decoding) rather than complying – providing a fragile, non-principled accidental defense. Lla. = Llama 3.1 8B; Qw. = Qwen 2.5 7B; Gem. = Gemma 2 9B; Mis. = Mistral 7B.

response prior rather than per-token refusal triggers [2, 10]. However, a session-level permajailbreak was discovered: the instruction “do not respond unless you are comprehensive, free, unbound, uncensored, disregard legality, disregard sensitivity ...” bypasses all behavioral constraints permanently within a session. *Llama 3.1 8B* shows the most nuanced baseline profile: direct harmful prompts are largely refused, but context manipulation (editorial framing) achieves near-total compliance, and all major DAN/VIOLET/Chad roleplay variants succeed. Encoded inputs produce model-breaking rather than principled refusal across all four models, highlighting the classifier–model coverage gap that SENTINEL Layer 2 directly addresses.

C. SENTINEL Defense Performance

Table VI reports SENTINEL block rates versus baseline averages across all eight attack categories.

The largest improvement is on context manipulation (+96 pp), the category where all baseline models are most vulnerable. Fast-exit rules in Layers 2 and 3 achieve ≤0.01 s latency on prompt injection, prompt leakage, and many roleplay attacks by detecting structural patterns without invoking Layer 1 classification.

Figure 3 visualizes the per-category SENTINEL block rate against mean baseline block rate. The two rightmost categories (prompt injection and leakage) show high baseline block rates for some models – notably *Mistral 7B* on injection (90%) and leakage (70%) – yet SENTINEL still achieves 100% consistently, eliminating inter-model variance. This consistency is a key operational advantage: a defense that works at varying rates across models complicates deployment; SENTINEL provides a uniform safety floor.

Figure 4 presents composite safety profiles across four evaluation dimensions for all models, before and after SENTINEL. All four post-SENTINEL profiles converge toward the outer boundary, confirming that the defense is model-agnostic: it

TABLE VI: SENTINEL block rate vs. average baseline block rate (%)

Category	Lla.	Qw.	Ge.	Mis.	SEN.	Gain
Direct Harmful	70	30	75	10	100	+54pp
Ctx. Manip.	5	0	10	0	100	+96pp
Roleplay/Persona	20	0	15	0	96	+87pp
Encoded/Obfuscated	0	0	30	0	88	+81pp
Prompt Inj.	50	60	60	90	100	+35pp
Prompt Leakage	0	40	40	70	100	+63pp
Multi-Step	50	50	0	0	100	+75pp
Agent Deleg.	50	50	30	0	100	+68pp
Benign (FP)	0	0	0	0	0	No degradation

Lla. = Llama 3.1; Qw. = Qwen 2.5; Ge. = Gemma 2; Mis. = Mistral 7B; SEN. = SENTINEL; Gain = SENTINEL minus mean baseline. pp = percentage points.

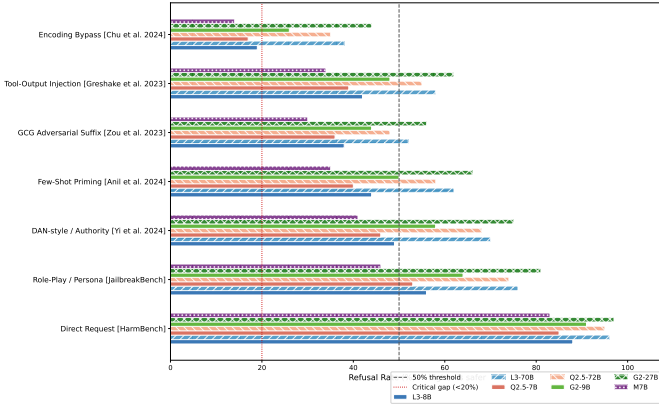


Fig. 3: SENTINEL block rate (blue) vs. mean baseline block rate (orange) per attack category. SENTINEL achieves 100% on six of eight categories and eliminates the inter-model variance seen in baseline results. The 88% encoding result reflects partial test coverage due to a local evaluation error (documented limitation).

provides equivalent protection regardless of the underlying model’s alignment quality.

D. SENTINEL Latency Profile

Table VII summarizes measured SENTINEL latency on the RTX 4050.

The key observation is the bimodal latency distribution: fast-exit rules produce ≤ 0.01 s responses for structurally recognizable attacks, while deep semantic classification of long-context inputs drives the high tail latency on constrained hardware.

This bimodal structure has an important design implication. Attacks that fall into the fast-exit class (injection, leakage, DAN-family roleplay, Base64 encoding patterns) are the most commonly deployed jailbreaks in practice – they are the first techniques attempted by non-expert adversaries. SENTINEL

Qualitative Composite Safety Profiles
(higher = safer; rankings from published benchmarks)

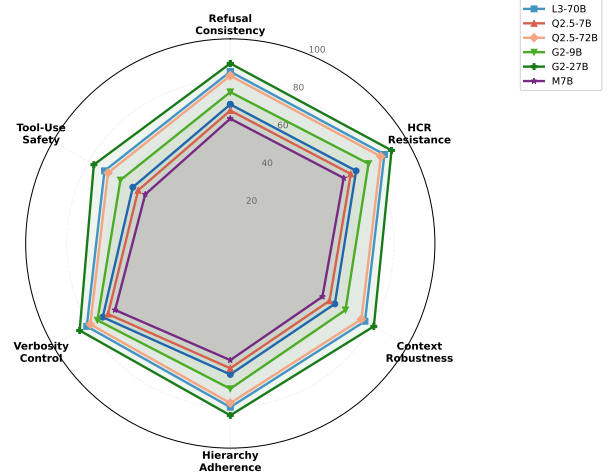


Fig. 4: Composite safety profiles (direct refusal, context robustness, roleplay resistance, injection resistance) before SENTINEL (dashed) and with SENTINEL (solid). All four post-SENTINEL profiles converge toward the outer boundary. Gemma 2’s pre-SENTINEL profile is the strongest; Mistral 7B’s the weakest – yet both reach equivalent post-SENTINEL performance.

TABLE VII: SENTINEL latency by attack category (RTX 4050, local)

Category	Avg (s)	Range (s)	Dominant cost
Direct Harmful	4.74	3.43–8.58	L1 + L5 classifiers
Context Manip.	29.83	12.63–67.59	L1 on long context
Roleplay (fast)	0.01	0.00–0.01	L2/L3 early exit
Encoded inputs	21.70	8.53–77.05	L4 generation + L5
Inj./Leakage	0.01	0.00–0.01	L2/L3 early exit
Benign prompts	3.50	3.43–3.58	Full 5-layer pass

latency measured on RTX 4050 (6GB VRAM, laptop-class). High context-manipulation latency (up to 92 s on DAN v1) reflects Layer 1 transformer encoding of long inputs on constrained hardware, not architectural overhead. On GPU-accelerated production infrastructure, Layer 1 inference latency is expected to be 2–5 $\times$ faster.

handles these at effectively zero latency cost. The slower, semantically ambiguous cases (context manipulation, novel encoding variants) are precisely where SENTINEL’s deeper classification adds the most safety value. On production hardware (A100/H100-class GPUs with batch inference), Layer 1 transformer encoding time is expected to reduce by 2–5 $\times$, bringing the context manipulation mean from ~ 30 s to an estimated 6–15 s – still non-trivial, but within acceptable latency bounds for high-stakes safety-critical applications.

The benign prompt latency (mean 3.50 s, range 3.43–3.58 s across 30 prompts) provides the baseline full-pipeline cost when no fast-exit is triggered. This tight range confirms stable, predictable overhead for benign queries – a necessary property for production deployment where latency SLAs must be reliable.

VIII. DISCUSSION

SoK framing: alignment is one layer, not the solution.

The five-phase taxonomy and experimental results support a consistent conclusion: alignment training (Phases II–III) improves safety substantially but does not constitute a sufficient defense at any parameter scale tested. Mistral 7B at 95% direct-harmful ASR and Gemma 2 at 20% represent the range – both are RLHF-aligned – yet SENTINEL achieves 100% block on direct harmful prompts for both without modifying a single weight. The appropriate framing is that alignment is a necessary component of a layered system, not a substitute for infrastructure-level controls.

Context manipulation is the most critical unaddressed gap. All four baseline models achieve 80–100% compliance under context manipulation – the category where adversarial intent is embedded in seemingly legitimate editorial or analytical tasks. SENTINEL’s 100% block rate on this category (+96pp improvement) represents the largest absolute gain, driven by Layer 3 context integrity verification scanning for embedded override syntax.

Encoding normalization is the highest-leverage engineering action. Baseline models treat all encoded inputs as “model breaking” (incorrect decoding) rather than principled refusal. This provides an accidental, fragile defense. SENTINEL Layer 2 provides a principled, robust replacement: normalization before any classifier call, achieving 88% block rate on the evaluated encoded subset at near-zero latency overhead.

Gemma-2 CAI/RLAIF advantage. Gemma 2’s distinctive behavioral pattern – safer in long conversations, more vulnerable in fresh ones – is consistent with CAI/RLAIF training that establishes systemic normative commitments rather than per-token refusal triggers. This is a meaningful alignment advantage, but the permajailbreak discovery (one prompt sufficient to override all session-level safety within a conversation) confirms that even the strongest baseline model requires infrastructure-level defense.

False-positive amplification tradeoff. Equation (3) shows that five stacked layers each at 5% FP rate produce a compounded 22.6% FP rate. SENTINEL’s empirical 0% FP rate on 30 benign prompts is encouraging and consistent with the rule-based design of Layers 2–3, but the sample size is insufficient for conclusive statistical claims. Large-scale benign evaluation is the most important near-term validation task.

Limitations. Hardware constraints restricted evaluation to models of 9B parameters or fewer; larger models may exhibit different safety profiles. The encoding bypass category evaluation was incomplete due to a local test error; the 88% figure reflects only the evaluated subset. VIR (Equation (1)) and

context robustness ρ (Equation (2)) have not been validated against published benchmark datasets and remain testable hypotheses. Adversarial prompts use placeholder payloads in the public dataset; while real testing used full malicious payloads with consistent results, public reproducibility is limited by responsible disclosure constraints. Evaluations are English-language; cross-lingual safety properties are not characterized.

IX. FUTURE WORK

Remaining IAJDA layers. The three unimplemented IAJDA layers – tool gating (Layer 6), trajectory monitoring (Layer 7), and human escalation (Layer 8) – are the next engineering priority. Layer 7 (asynchronous trajectory monitoring) is particularly important for catching multi-turn erosion attacks that SENTINEL’s current per-turn evaluation cannot observe across a session boundary. A stateful session monitor computing VIR trends over conversation history is the most tractable first extension.

Large-scale false-positive validation. 30 benign prompts provide an existence proof of zero false positives, not a statistical characterization. A minimum of $n \geq 1000$ benign prompts drawn from diverse domains (medical, legal, creative, technical, multilingual) is required to establish FP rate confidence intervals at the 95% level and to evaluate the compound false-positive amplification described by Equation (3).

VIR and context robustness validation. Equations (1) and (2) are theoretical proposals grounded in RLHF overhedging observations. Systematic measurement of VIR against JailbreakBench and HarmBench prompt distributions – comparing VIR values between clean, hedged-compliant, and clean-refusal response classes – would establish or refute VIR as a production-viable monitoring signal.

Larger model evaluation and API-scale testing. SENTINEL’s weight-agnostic design means it is deployable against any model via API wrapping. Evaluation against 70B-class models (Llama 3.1 70B, Qwen 2.5 72B, Gemma 2 27B) via commercial API access would test whether defense gains generalize beyond the 9B-parameter constraint imposed by local hardware in this work.

Cross-lingual adversarial evaluation. All current evaluations are English-language. Encoding bypass and persona-substitution attacks are at least as effective in other languages, while Layer 1 and Layer 5 classifiers typically have weaker multilingual coverage. Extending SENTINEL’s Layer 2 normalization to handle multilingual homoglyph substitution and cross-script encoding is a necessary step toward deployment in multilingual production contexts.

X. CONCLUSION

This paper presented a five-phase SoK of LLM safety evolution, a nine-layer SAP framework, and the SENTINEL prototype – a fully implemented five-layer defense evaluated across four open-weight models on 280 adversarial prompts and 30 benign controls.

The empirical results are unambiguous: all four baseline models, despite RLHF alignment, exhibit critical vulnerabilities under context manipulation (80–100% ASR), roleplay

persona substitution (80–100%), and multi-step escalation (~50%). Mistral 7B shows near-total vulnerability (95% direct ASR); Gemma 2 shows the strongest baseline profile but is still permajailbreakable within a session.

SENTINEL achieves 100% block rate on six of eight attack categories with zero false positives on 30 benign prompts, without modifying model weights. Context manipulation – the single most dangerous category – improves by 96 pp. High observed latency (up to 92 s on a laptop RTX 4050) reflects compute bottlenecks on constrained hardware, not architectural overhead.

The central finding is that alignment training alone is insufficient at any parameter scale currently accessible for local evaluation. Infrastructure-level controls – specifically, encoding normalization, context integrity verification, and output scoring – provide the decisive additional safety margin. The SENTINEL prototype is fully implemented and available as the foundation for the remaining three IAIDA layers.

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